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1.

```
class Solution {
public:
    ListNode* middleNode(ListNode* head) {
        ListNode* slow = head;
        ListNode* fast = head;

        while(fast != NULL && fast->next != NULL) {
            slow = slow->next;
            fast = fast->next->next;
        }

        return slow;
    }
};
```

1.

```
class Solution {
public:
    bool hasCycle(ListNode *head) {
        ListNode* slow = head;
        ListNode* fast = head;

        while(fast != NULL && fast->next != NULL) {
            slow = slow->next;
            fast = fast->next->next;

            if(slow == fast)
```

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        return true;
    }

    return false;
}
};
```

2.

```
class Solution {
public:
    ListNode *detectCycle(ListNode *head) {
        ListNode* slow = head;
        ListNode* fast = head;

        // Step 1: detect cycle
        while(fast != NULL && fast->next != NULL) {
            slow = slow->next;
            fast = fast->next->next;

            if(slow == fast) {
                // Step 2: find start
                ListNode* temp = head;

                while(temp != slow) {
                    temp = temp->next;
                    slow = slow->next;
                }

                return temp;
            }
        }

        return temp;
    }
};
```

```
    }  
}  
  
return NULL;  
}  
};
```